Figures

Density of human hair [1] = 1.34 g/cm^3 = 1340 kg/m^3

Diameter of average human hair [2] = 75 µm

Average number of hairs on human head [3] = 100,000

Average growth of human hair every 28 days [4] = 1cm = 0.01 m

Global urban population by 2030 [5] = 5.2 billion people

Global yearly wool yield [6] = 1,160,000,000 kg = 1,160,000 tonnes

Equations

$r = D/2$

$A = \pi r^2$

$\rho = \frac{m}{V}$

Calculations

Area of human hair cross section = πr^2 = $\pi(3.75 \times 10^{-5})^2$ = 4.42×10^{-9} m^2

Mass of human hair per unit length = ρA = 1340 x 4.42 x 10^{-9} = 5.92 x 10^{-6} kg/m

Radius of human hair = $D/2$ = $\frac{75}{2}$ = 37.5 µm = 3.75x10^{-5} m

Average yearly growth of human hair = $\frac{number\ of\ days\ in\ year}{28}$ x 28 days growth rate = $\frac{365 \times 0.01}{28}$ = 0.13 m

Total mass of hair produced by one person in a year = number of hairs x yearly growth rate x mass per unit length = 100000 x 0.13 x 5.92 x 10^{-6} = 0.077 kg

Total mass of hair proceed by the urban population of the planet in 2030 = urban population x yearly hair

Yield Per Person = 5200000000 x 0.077 = 400,000,000 kg = 400,000 tonnes

Yearly human hair yield as a fraction of yearly wool yield = Yearly wool yield/ yearly human hair yield = $\frac{100 \times 400000}{1160000}$ = 34%

References

[1] - K., Eshwara & P., Divakara & C., Udaya. (2018). Experimental Studies on Behavior of Keratin Based Human Hair Fiber - A New Reinforcing Material for Composites. International Journal of Engineering & Technology. 7. 459-463. 10.14419/ijet.v7i4.5.20207.

[2] - “Hair’s Breadth.” *Wikipedia*, Wikimedia Foundation, 15 May 2022, en.wikipedia.org/wiki/Hair%27s_breadth.

[3] - BNID 101509, Milo et al. Nucl. Acids Res. (2010) 38 (suppl 1): D750-D753

[4] - BNID 109909, Milo et al. Nucl. Acids Res. (2010) 38 (suppl 1): D750-D753

[5] - United Nations, Department of Economic and Social Affairs, Population Division (2018). World Urbanization Prospects: The 2018 Revision, Online Edition. (link)

[6] - International Wool Textile Organization (2017) Wool Industry, accessed 2 August 2017